EV Charger



Category: **Automotive** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: AC EV Charger (Slow); DC fast Charger; DC Fast Charger (Demo Unit); EV Charging Gun (Type-2)

What it is used for

An EV charger supplies electrical energy to a compatible electric vehicle through an approved charging connection. Correct equipment matching, inspection, and charging procedures help maintain reliable charging and reduce electrical hazards.

Main parts

- Charging unit controls and monitors the charging process.
- Charging connector or gun connects the charger to the vehicle inlet.
- Charging cable carries electrical power and communication signals as designed.
- Control and protection circuits monitor faults and charging conditions.
- Display or status indicators show charging state on equipped models.
- AC chargers provide alternating current for the vehicle's onboard charger.
- DC fast chargers supply controlled DC power through compatible vehicle systems.
- Emergency stop and protective devices are provided on applicable charging units.

Before you start

1. Read the charger and vehicle manuals and confirm connector and charging compatibility.
2. Check applicable site rules, charger rating, and required user authorisation.
3. Inspect the charger, cable, connector, plug, and vehicle inlet for damage or contamination.
4. Ensure connectors and hands are dry and the charging area is clear.
5. Check that the charger is free from warning indicators or visible faults.
6. Confirm the vehicle is parked safely and switched off or placed in the required charging state.

How to use

1. Position the vehicle within reach of the charging cable without stretching or crushing it.
2. Activate the charger using the approved access method, if required.
3. Open the vehicle charging port and inspect it for visible debris or damage.
4. Connect the correct charging connector firmly to the vehicle inlet as instructed.
5. Follow the charger's displayed prompts or approved procedure to start the session.
6. Check the display or vehicle indicators to confirm charging has started normally.
7. Monitor charging status and respond to warnings according to the manual.
8. To finish, stop the session using the approved controls before disconnecting the connector.
9. Remove the connector by its handle without pulling the cable and return it to its holder.
10. Close the vehicle charging port, secure the cable, and report any faults.

⚠ Safety

- Never use a charger, cable, or connector with visible damage or exposed conductors.
- Do not touch connectors with wet hands or use them in standing water.
- Never force a connector into the vehicle inlet or modify the plug.
- Do not disconnect the charging connector under load unless the equipment procedure explicitly permits it.
- Do not open charger enclosures or attempt electrical repairs without authorised training.
- If there is smoke, unusual heat, sparking, or an electric shock risk, stop if safe, keep away, and contact emergency or authorised service personnel.

Care & storage

- Keep connectors and cable holders clean and protected from impact.
- Inspect cables, connectors, and visible enclosure parts regularly for wear or damage.
- Keep the charging area clear and follow the manufacturer's cleaning instructions.
- Report repeated charging failures, warning indicators, or damaged equipment for authorised inspection.

Common mistakes to avoid

- Using a connector or charging mode incompatible with the vehicle.
- Starting a session without inspecting the cable and connectors.
- Pulling the cable instead of releasing the connector by its handle.
- Disconnecting before stopping the charging session as instructed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar